Symmetric Matrix

Tuesday, 18 March, 2025 9:35 AM

$A = A^T$

$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{matrix} \nearrow x_i \\ \searrow \lambda_i \end{matrix}$

1) eigen values are always real.

2) eigen vectors are always perpendicular.

3) eigen vectors can be chosen to be orthonormal.

$A = S \Lambda S^{-1}$

$S = \begin{bmatrix} \vdots & \vdots & \vdots & & \\ x_1 & x_2 & x_3 & \cdots & n \\ \vdots & \vdots & \vdots & & \end{bmatrix}$

$Q = \begin{bmatrix} \vdots & \vdots & \vdots & & \\ q_1 & q_2 & q_3 & \cdots & n \\ \vdots & \vdots & \vdots & & \end{bmatrix}$

$A = Q \Lambda Q^{-1}$

$= Q \Lambda Q^T$

$$A = \underbrace{\begin{bmatrix} \vdots & \vdots & \vdots \\ q_1 & q_2 & q \cdots n \\ \vdots & \vdots & \vdots \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix}}_{\Lambda} \underbrace{\begin{bmatrix} \cdots & q_1 & \cdots \\ \cdots & q_2 & \cdots \\ \cdots & q_3 & \cdots \\ & q_n & \end{bmatrix}}_{Q^T}$$

$$= \lambda_1 \underbrace{q_1 q_1^T}_{P1} + \lambda_2 \underbrace{q_2 q_2^T}_{P_2} + \cdots\cdots + \lambda_n \underbrace{q_n q_m^T}_{P_n}$$

$a^T a = \|a\|^2$

$P = \dfrac{q_i q_i^T}{q_i^T q_i}$

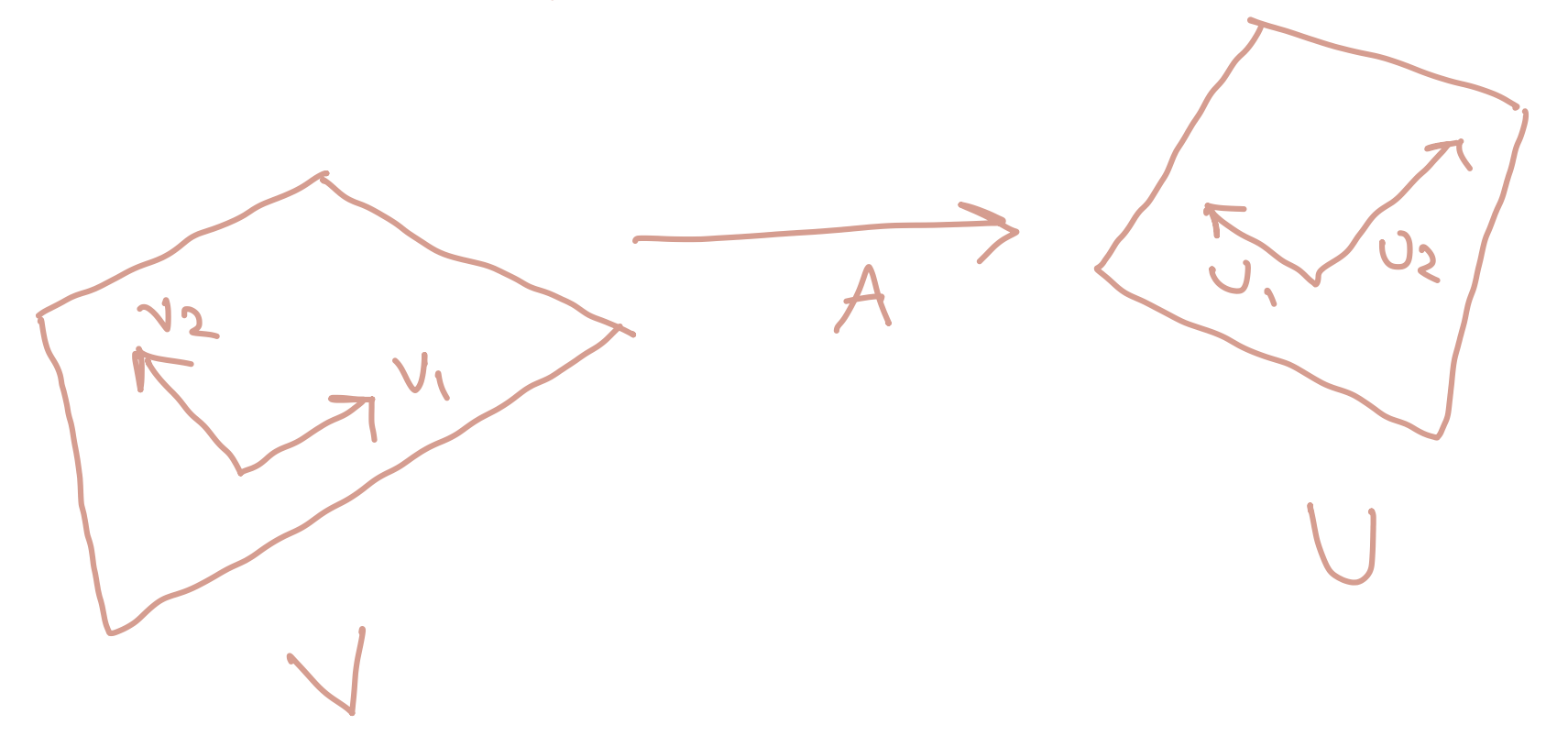
$P = \dfrac{q_i q_i^T}{1}$

Singular value Decomposition (SVD)

$A = LDU$

$A = S \Lambda S^{-1}$

$A = U \Sigma U^T$



$A v_1 = \sigma_1 u_1$

$\vdots$

$A v_r = \sigma_r u_r$

$A \rightarrow m \times n; \quad m \neq n$

$A^T A \rightarrow$ square symmetric $\longleftarrow A A^T$

$A^T A = V \Lambda V^{-1}$

$A A^T = U \Lambda U^T$

$A v_i = \sigma_i u_i$

$$\underbrace{\begin{bmatrix} & \\ & \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} \vdots & \vdots & \vdots & & \vdots \\ v_1 & v_2 & v_3 & \cdots & v_r \\ \vdots & \vdots & \vdots & & \vdots \end{bmatrix}}_{V} = \underbrace{\begin{bmatrix} \vdots & \vdots & & \vdots \\ u_1 & u_2 & \cdots & u_r \\ \vdots & \vdots & & \vdots \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{bmatrix}}_{\Sigma}$$

$AV = U\Sigma$

$A = U \Sigma V^T$

$A^T A = (U \Sigma V^T)(U \Sigma V^T)$

$= V (\Sigma^T \Sigma) V^T$

$$\underbrace{\begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{bmatrix}}_{\Sigma^T} \underbrace{\begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{bmatrix}}_{\Sigma}$$

$$\underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}}_{\Lambda} = \underbrace{\begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \sigma_3^2 & \\ & & & \ddots \; \sigma_r^2 \end{bmatrix}}_{\Sigma^T \Sigma}$$

$\lambda_i = \sigma_i^2$

$\sigma_i = \sqrt{\lambda_i}$

# $A = \begin{bmatrix} 5 & 5 \\ -1 & 7 \end{bmatrix}$

$A = U \Sigma V^T$ | $A^T A = V (\Sigma^T \Sigma) V^T$

$AV = U\Sigma$

$A^T = \begin{bmatrix} 5 & -1 \\ 5 & 7 \end{bmatrix}$

$A^T A = \begin{bmatrix} 26 & 18 \\ 18 & 74 \end{bmatrix}$

$A^T A = \begin{bmatrix} 26 & 18 \\ 14 & 74 \end{bmatrix}$ $\begin{vmatrix} 26-\lambda & 18 \\ 18 & 74-\lambda \end{vmatrix} = 0$

$\lambda^2 - 100\lambda + 1600 = 0$

$\lambda_1 = 80 \quad \lambda_2 = 20$

$\begin{bmatrix} 6 & 18 \\ 18 & 54 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \rightarrow \begin{bmatrix} \frac{3}{\sqrt{10}} \\ -\frac{1}{\sqrt{10}} \end{bmatrix} \rightarrow v_1$

$\begin{bmatrix} 1 \\ 3 \end{bmatrix} \rightarrow \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix} \rightarrow v_2$

$\sigma_i = \sqrt{\lambda_i}$

$\Sigma = \begin{bmatrix} \sqrt{20} & 0 \\ 0 & \sqrt{80} \end{bmatrix}$

$\begin{bmatrix} 5 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{-1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{bmatrix}$

$= \begin{bmatrix} -\sqrt{10} & 2\sqrt{10} \\ \sqrt{10} & 2\sqrt{10} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

$= U \Sigma \qquad = U$

$\Sigma = \begin{bmatrix} \sqrt{20} & 0 \\ 0 & \sqrt{80} \end{bmatrix}$

(Check if the normalized or not)